Piper PA-31-310 Navajo (NAVAJO)

2× Lycoming TIO-540-A2C turbocharged, 310 hp each at 2575 RPM / 42 in Hg · 6–8 seats · Typical PA-31-310 Navajo (6,500 lb MTOW, unpressurised) POH procedures; hydraulic gear and flaps, inboard/outboard tanks per side, unfeathering accumulators and emergency gear extension (CO2 or hand pump) vary by serial – verify your aircraft.

V-speeds & limits

Vr	**85–90 KIAS (rotate / liftoff)**
Vmca	**74 KIAS (red line)**
Vsse	**85 KIAS approx (intentional one-engine-inop)**
Vx	**83 KIAS**
Vxse	**90 KIAS**
Vyse	**100 KIAS (blue line)**
Vy	**107 KIAS**
Best glide	**approx 105–115 KIAS (both engines out, props feathered – verify POH)**
Vfe	**155 KIAS (15°) / 125 KIAS (40°)**
Vle	**156 KIAS (gear extended / extension)**
Vlo	**130 KIAS (retraction)**
Va	**155 KIAS (6,500 lb)**
Vno	**180 KIAS**
Vne	**236 KIAS**
Vs / Vso	**approx 72 / 62 KIAS (max weight, verify)**
Vref	**95–100 KIAS final (flaps 40°); 100 KIAS single-engine until landing assured**

Normal procedures

Before Start

Preflight inspection	**COMPLETE (fuel caps, drains, turbo exhaust / tailpipes, gear doors, oil 8–12 qt)**
Fuel quantity	**CHECK visually – inboard 56 gal and outboard 40 gal per side (approx 187 gal usable total)**
Nose baggage door / wing locker doors	**CLOSED & LOCKED**
Passenger briefing	**COMPLETE (belts, airstair door, emergency exit, no smoking, sterile cockpit)**
Seats, belts, shoulder harnesses	**ADJUST & LOCK**
Airstair cabin door	**CLOSED & LATCHED (upper and lower halves, handle flush, pins visible)**
Parking brake	**SET**
Landing gear selector	**DOWN**
Avionics master	**OFF**
All electrical switches	**OFF**
Circuit breakers	**CHECK IN**
Alternate static source	**OFF (normal)**
Fuel selectors	**INBOARD (both – takeoff and landing on inboard tanks)**
Crossfeed	**OFF**
Cowl flaps	**OPEN**
Alternator switches	**ON (both)**
Throttles / propellers / mixtures	**IDENTIFY – throttles closed, props full forward, mixtures idle cutoff**
Cabin heater	**OFF**

Engine Start

Battery master	**ON**
Gear lights	**THREE GREEN**
Annunciators	**PRESS TO TEST**
Beacon	**ON**
Propeller area	**CLEAR (both sides)**
Mixture	**FULL RICH**
Propeller	**FULL FORWARD (high RPM)**
Throttle	**OPEN approx 1/2 inch**
Electric fuel pump	**ON until fuel flow indicates, then OFF (prime)**

▶ Engine hot or flooded?

No:

Yes:

Mixture	**IDLE CUTOFF**
Throttle	**FULL OPEN**
Mixture	**ADVANCE to FULL RICH as engine fires, throttle to idle**
Magneto / starter switch	**START (10 s max, then 30 s cool-down)**
Throttle	**800–1000 RPM – avoid rapid throttle movement with a cold turbocharger**
Oil pressure	**25 psi within 30 s, rising to 60–90 psi (100 psi max cold)**
Second engine	**REPEAT start sequence**
Alternators / ammeters	**CHECK both charging, voltage 27–29 V**
Electric fuel pumps	**OFF (fuel flow steady on engine-driven pumps)**
External power	**DISCONNECTED (if used)**
Avionics master	**ON**
Hydraulic pressure	**CHECK in green**
Engine instruments	**CHECK both (oil pressure, oil temp rising, fuel flow, TIT, CHT)**

Before Taxi

Flight instruments / avionics / GPS	**CHECK & SET**
Altimeters	**SET**
Heading indicator / HSI	**SET to compass**
Autopilot / electric trim	**CHECK & OFF**
Transponder	**STBY / as required**
Parking brake	**RELEASE**
Brakes	**CHECK on initial roll**
Nosewheel steering	**CHECK (rudder pedals)**
Gyros	**CHECK during turns (turn coordinator, DG, attitude indicator)**
Fuel selectors	**INBOARD both, crossfeed OFF**
Cowl flaps	**OPEN**

Before Takeoff

Parking brake	**SET (into wind, oil temp above 75°F / 24°C before run-up)**
Flight controls	**FREE & CORRECT**
Trims (elevator, rudder, aileron)	**SET for takeoff**
Fuel selectors	**INBOARD both (feel for detents), crossfeed OFF**
Mixtures	**FULL RICH**
Propellers	**FULL FORWARD**
Throttles	**2000 RPM (one engine at a time)**
Magnetos	**CHECK L & R each engine (175 RPM max drop, 50 RPM max difference)**
Propellers	**CYCLE 2–3 times to low RPM (RPM drops approx 300–500, returns)**
Propellers – feather check	**1500 RPM, pull through detent to FEATHER – RPM drops; return to full forward before 1000 RPM**
Unfeathering accumulators	**CHECK charged (if fitted – allow pressure to rebuild after feather check)**
Alternators	**CHECK both charging under load**
Gyro suction / vacuum	**CHECK 4.8–5.1 in Hg, both pump annunciators out**
Engine instruments	**CHECK both (oil pressure 60–90 psi, oil temp green, CHT, TIT, fuel flow)**
Hydraulic pressure	**CHECK in green**
Throttles	**IDLE check (600–700 RPM), then 1000 RPM**
Electric fuel pumps	**ON (both)**
Cowl flaps	**OPEN**
Propellers	**FULL FORWARD**
Flaps	**UP or 15° (short field) – check indicator**
Landing gear selector	**DOWN – three green**
Pitot heat / de-ice	**AS REQUIRED**
Lights	**AS REQUIRED**
Transponder	**ALT**
Airstair door / windows	**LATCHED (door annunciator out, if fitted)**
Takeoff briefing	**COMPLETE (Vmca 74 / Vr 85–90 / Vyse 100; abort below liftoff; engine-out plan)**
Takeoff power	**ADVANCE SMOOTHLY – 42 in Hg / 2575 RPM max, retard if MP overshoots (do not overboost)**

Turbocharger response lags the throttle. Advance throttles slowly over 3–5 seconds and check manifold pressure before brake release; set final power early in the roll.

After Takeoff / Climb

Landing gear	**UP (positive rate, no usable runway, below 130 KIAS)**
Gear lights	**OUT (three green out, no red in-transit light)**
Flaps	**UP (if used, above 90 KIAS)**
Climb power	**REDUCE SMOOTHLY – 2400 RPM / approx 35 in Hg (75%) after obstacles**
Airspeed	**107 KIAS (Vy) / 120–130 KIAS cruise climb**
Mixtures	**FULL RICH (fuel flow in green; lean only per POH climb schedule, TIT below 1650°F)**
Electric fuel pumps	**OFF above approx 1000 ft AGL (check fuel flow steady)**
Cowl flaps	**OPEN (CHT in green (approx 475°F max), oil temp below 245°F)**
Engine instruments	**CHECK both (MP, RPM, fuel flow, TIT, CHT, oil)**
Landing / taxi lights	**OFF**

Cruise

Cruise power	**SET smoothly – 2200–2400 RPM / 30–34 in Hg typical (set MP after RPM)**
Mixtures	**LEAN to TIT per POH cruise table (1650°F max, typically 100°F rich of peak)**
Cowl flaps	**CLOSED (as required for CHT)**
Propellers	**SYNCHRONISE**
Fuel selectors	**OUTBOARD (level cruise only) – electric fuel pumps ON during tank change, OFF when flow steady**
Fuel quantity / flow	**CHECK against plan; return to INBOARD before outboards run dry and for descent**
Crossfeed	**OFF (one-engine-inop balance only)**
Engine instruments	**CHECK both (TIT, CHT, oil temp 245°F max, oil pressure 60–90 psi)**
Altimeter / heading	**CHECK**
Cabin heater / oxygen	**AS REQUIRED**

Descent

Altimeters	**SET**
Fuel selectors	**INBOARD both – electric fuel pumps ON during change, then OFF**
Power	**REDUCE GRADUALLY (approx 2 in Hg per minute – avoid shock cooling; keep MP above gear horn setting)**
Mixtures	**ENRICH progressively**
Cowl flaps	**CLOSED**
Airspeed	**PLAN below 156 KIAS for gear, below 155 KIAS for 15° flap**
Approach briefing	**COMPLETE (incl. single-engine go-around decision point)**
Seat belts / harnesses	**SECURE – passengers briefed**

Before Landing

Fuel selectors	**INBOARD both, crossfeed OFF**
Electric fuel pumps	**ON (both)**
Mixtures	**FULL RICH**
Cowl flaps	**AS REQUIRED**
Airspeed	**BELOW 156 KIAS**
Landing gear	**DOWN – THREE GREEN, hydraulic pressure recovered**
Flaps	**15° below 155 KIAS; 40° on final below 125 KIAS**
Propellers	**FULL FORWARD (on final)**
Airspeed	**95–100 KIAS final; approx 90 KIAS over threshold**
Landing lights	**ON**
Landing gear	**THREE GREEN – CONFIRMED on final**

After Landing

Flaps	**UP (identify flap lever before moving)**
Cowl flaps	**OPEN**
Electric fuel pumps	**OFF**
Pitot heat / de-ice	**OFF**
Landing lights / strobes	**AS REQUIRED**
Transponder	**STBY / as required**
Trims	**SET takeoff**

Shutdown

Parking brake	**SET**
Throttles	**1000 RPM for 3–5 min (turbocharger cool-down before shutdown)**
Avionics master	**OFF**
Electrical equipment / lights	**OFF**
Mixtures	**IDLE CUTOFF**
Magnetos	**OFF (after propellers stop)**
Alternator switches / battery master	**OFF**
Fuel selectors	**OFF (if required by operator) or INBOARD**
Control lock	**INSTALL**
Chocks / tiedowns / pitot covers	**AS REQUIRED**

Abnormal procedures

Alternator Failure / Low Voltage

ALT annunciator, one ammeter zero or voltage below approx 26 V

Affected alternator switch ..... **OFF, then ON (reset)**

Alternator field circuit breaker ..... **CHECK / RESET once**

▶ Alternator restored?

Yes:

Ammeters / voltmeter ..... **MONITOR both**

No – one alternator failed:

Affected alternator switch ..... **OFF**

Electrical load ..... **REDUCE to within remaining alternator capacity (70 A)**

Deferred: One alternator inoperative – electrical load reduced, land as soon as practical. ..... **NOTED**

No – both failed:

Both alternator switches ..... **OFF**

Non-essential electrical load ..... **REDUCE (battery only, limited endurance; avionics minimum)**

Landing ..... **PLAN early – gear is hydraulic but gear lights, electric fuel pumps and flap indication need battery power**

Deferred: Both alternators failed – battery only; land as soon as possible, confirm gear by hydraulic pressure cycle. ..... **NOTED**

Low Oil Pressure (one engine)

Oil pressure below 60 psi in flight or below 25 psi at idle

Oil pressure / oil temperature ..... **CHECK and cross-check the affected engine**

▶ Oil temperature rising or pressure below 25 psi?

Yes – genuine:

Affected engine power ..... **REDUCE to minimum required**

Engine ..... **SHUT DOWN and FEATHER before seizure if altitude and terrain allow (Engine Failure in Flight)**

Feathering needs oil pressure – feather early while the engine still turns.

Deferred: Low oil pressure – precautionary engine shutdown; single-engine landing, land as soon as possible. ..... **NOTED**

No – gauge suspect:

Flight ..... **CONTINUE, monitor oil temperature and pressure**

Deferred: Oil pressure indication suspect – maintenance check required. ..... **NOTED**

Turbocharger Failure / Manifold Pressure Loss

Manifold pressure drops toward ambient and cannot be restored; possible exhaust leak, smoke or roughness

Affected engine throttle ..... **REDUCE, then re-set smoothly – do not pump the throttle**

Affected engine mixture ..... **LEAN to fuel flow appropriate to the reduced power (TIT / EGT rises as mixture richens on a dead turbo)**

Engine instruments ..... **CHECK oil pressure, oil temp, TIT, CHT, fuel flow**

▶ Engine smooth with normally-aspirated power, no smoke or fire indication?

Yes:

Power ..... **OPERATE at the reduced power available; mixture as required**

Altitude ..... **DESCEND as required – single-engine ceiling and cruise altitude reduced**

Deferred: Turbocharger inoperative one side – reduced power and asymmetric thrust at altitude, land as soon as practical. ..... **NOTED**

No – roughness, smoke or fluctuating oil pressure:

Affected engine ..... **SHUT DOWN and FEATHER (Engine Failure in Flight) – exhaust leak may cause nacelle fire**

Deferred: Engine shut down for turbo / exhaust failure – single-engine landing, land as soon as possible. ..... **NOTED**

High TIT / Overboost

TIT above 1650°F or manifold pressure above 42 in Hg

Affected engine throttle ..... **RETARD to bring MP below 42 in Hg**

Affected engine mixture ..... **ENRICH until TIT below 1650°F**

Affected engine cowl flap ..... **OPEN**

Airspeed ..... **INCREASE if CHT also high**

▶ TIT and MP within limits?

Yes:

Power / mixture ..... **RE-SET slowly; note any overboost (duration and MP) for maintenance**

Deferred: Overboost or TIT exceedance – record for maintenance entry. ..... **NOTED**

No:

Affected engine power ..... **REDUCE further – consider shutdown if TIT uncontrollable (Engine Failure in Flight)**

Deferred: TIT / MP uncontrollable – wastegate or controller fault, land as soon as practical. ..... **NOTED**

Fuel Flow Fluctuation / Vapour / Rough Running

Affected engine electric fuel pump ..... **ON**

Fuel selector ..... **CHECK INBOARD (or tank with fuel) – outboard tanks unport in turns or with low fuel**

Affected engine mixture ..... **ADJUST for smooth running**

Affected engine magnetos ..... **CHECK L / R individually, then BOTH**

Engine instruments ..... **CHECK (fuel flow, TIT, CHT, MP)**

▶ Engine running smoothly?

Yes:

Electric fuel pump ..... **LEAVE ON for the remainder of flight**

Deferred: Fuel flow fluctuation – electric pump on, inboard tanks; report to maintenance. ..... **NOTED**

No:

Crossfeed ..... **ON from other side if fuel starvation suspected**

Affected engine power ..... **REDUCE to smoothest setting – feather if power unusable (Engine Failure in Flight)**

Deferred: Engine rough – land as soon as practical at nearest suitable airport. ..... **NOTED**

Crossfeed Operation

Fuel balance with one engine inoperative, or fuel starvation suspected on one side

Flight attitude ..... **LEVEL (crossfeed is for level flight only)**

Fuel selector – operating engine ..... **INBOARD or OUTBOARD as required to select the tank supplying**

Fuel selector – inoperative engine side ..... **ON tank to be used (INBOARD or OUTBOARD)**

Crossfeed ..... **ON**

Electric fuel pumps ..... **ON during changeover, then as required**

Fuel flow / quantity ..... **MONITOR both sides**

Crossfeed ..... **OFF before approach – operating engine on its own INBOARD tank for landing**

Do not take off or land with crossfeed ON.

Deferred: Crossfeed used in flight – confirm crossfeed OFF and inboard tank selected before landing. ..... **NOTED**

Propeller Overspeed

RPM above 2575 and not responding to the prop lever

Affected engine throttle ..... **REDUCE**

Affected engine propeller ..... **LOW RPM (toward feather detent)**

Airspeed ..... **REDUCE to bring RPM within limits**

Affected engine oil pressure ..... **CHECK (governor is oil-operated)**

▶ RPM controllable at or below 2575?

Yes:

Power ..... **SET minimum that keeps RPM in limits**

Deferred: Propeller governor fault – control RPM with throttle and airspeed, land as soon as practical. ..... **NOTED**

No:

Affected engine ..... **SHUT DOWN and FEATHER (Engine Failure in Flight)**

Deferred: Engine shut down for prop overspeed – single-engine landing, land as soon as possible. ..... **NOTED**

Flap Failure / Asymmetry

Flap lever ..... **RETURN to current position – do not continue extension**

Hydraulic pressure ..... **CHECK (flaps and gear share the hydraulic system)**

▶ Flaps operable and symmetric?

Yes:

Flap setting ..... **AS REQUIRED**

No:

Landing ..... **AT CURRENT FLAP SETTING – flaps up approach approx 105–110 KIAS, longer runway**

Deferred: Flaps inoperative / asymmetric – flaps-up or partial-flap landing, increased landing distance; check gear extends normally. ..... **NOTED**

Gear Unsafe / Fails to Retract or Extend Normally

Red in-transit light stays on, fewer than three green, or gear warning horn

Airspeed ..... **BELOW 156 KIAS (below 130 KIAS for retraction)**

Hydraulic pressure ..... **CHECK – allow time for pump to complete cycle**

Gear light bulbs / navigation lights dimmer ..... **CHECK (bulb or dimming, swap bulbs)**

Landing gear selector ..... **RECYCLE once to the desired position**

▶ Gear in desired position with correct indication?

Yes:

Gear indication ..... **CONFIRM**

Deferred: Gear system fault – do not retract after landing; maintenance inspection required. ..... **NOTED**

No – failed to retract:

Landing gear selector ..... **DOWN – leave down**

Airspeed ..... **BELOW 156 KIAS**

Deferred: Gear will not retract – gear down, reduced performance and range (negative single-engine climb), land as soon as practical. ..... **NOTED**

No – failed to extend / unsafe:

Use Emergency Gear Extension. Request a visual check from ATC or another aircraft if available.

Airstair Cabin Door Open in Flight

DOOR AJAR annunciator (if fitted), noise or draught from the aft cabin

Aircraft ..... **CONTROLLABLE – fly the airplane**

Airspeed ..... **REDUCE to approx 110–120 KIAS**

Belts / loose items / passengers ..... **SECURE, passengers away from the door**

Do not attempt to close the airstair door in flight. The lower (stair) half may trail open; expect noise and buffet.

Deferred: Cabin door unlatched – return and land as soon as practical, normal approach speed. ..... **NOTED**

Vacuum / Gyro Failure

Vacuum annunciator on, suction below 4.8 in Hg, or attitude / heading indicator drifting

Suction gauge / pump annunciators ..... **CHECK (one engine-driven pump failed – remaining pump maintains system)**

Attitude / heading indicators ..... **CROSS-CHECK with turn coordinator, compass, GPS, standby instruments**

Failed instruments ..... **COVER**

Autopilot ..... **OFF if coupled to failed gyro**

IMC ..... **EXIT / AVOID if both pumps or the attitude indicator failed**

Deferred: Vacuum / gyro failure – partial panel; prefer VMC approach. ..... **NOTED**

Inadvertent Icing Encounter

Ice accumulating on wings, windshield or props

Pitot heat / stall warning heat ..... **ON**

Propeller de-ice / windshield heat ..... **ON (if fitted)**

Wing / tail de-ice boots ..... **CYCLE when approx 1/2 inch of ice has formed (if fitted)**

Alternate air ..... **CHECK – induction icing shows as MP loss; alternate air is automatic on the TIO-540**

Power ..... **INCREASE as required; cowl flaps as required**

Altitude / heading ..... **CHANGE to exit icing conditions (ask ATC)**

Autopilot ..... **OFF – hand fly, monitor airspeed**

▶ Aircraft certified and equipped for flight in known icing?

Yes:

Approach ..... **FLAPS 15° max, speed +10–15 KIAS, no slow flight**

Deferred: Ice accretion – partial-flap landing, approach speed increased, longer runway. ..... **NOTED**

No:

Icing conditions ..... **EXIT IMMEDIATELY**

Deferred: Icing encounter without approved equipment – flaps 15° max, approach speed increased, land as soon as practical. ..... **NOTED**

Pitot / Static Blockage

Airspeed, altimeter or VSI erratic or frozen

Pitot heat ..... **ON**

Alternate static source ..... **ON (pilot subpanel – expect small altimeter and airspeed changes)**

Cabin / storm windows / vents ..... **CLOSED when using alternate static**

Attitude / power / GPS groundspeed ..... **USE for pitch and speed reference**

▶ Indications restored?

Yes:

Deferred: Alternate static in use – apply POH correction, pitot / static drain and inspection required. ..... **NOTED**

No:

Approach ..... **FLY by attitude and power (known settings), VMC preferred, longer runway**

Deferred: Pitot / static blocked – unreliable airspeed, fly known power and attitude settings. ..... **NOTED**

Emergency procedures

Engine Failure During Takeoff

On the runway or just airborne with runway remaining

▶ Airborne?

No – below liftoff:

Throttles (memory) ..... **CLOSED (both)**

Brakes (memory) ..... **MAXIMUM – stop straight ahead**

Mixtures ..... **IDLE CUTOFF**

Fuel selectors ..... **OFF**

Magnetos ..... **OFF**

Battery master / alternators ..... **OFF**

Yes – runway remaining:

Throttles (memory) ..... **CLOSED (both)**

Land (memory) ..... **STRAIGHT AHEAD, gear down**

Brakes ..... **MAXIMUM**

Mixtures / fuel selectors / magnetos ..... **OFF after stop**

Yes – no runway remaining:

Go to Engine Failure After Liftoff.

Below Vmca (74 KIAS) directional control cannot be maintained with full power on the operating engine. Do not attempt to continue a takeoff below liftoff speed – the left engine is critical.

Engine Failure After Liftoff

Airborne, insufficient runway to land ahead

LAND AS SOON AS POSSIBLE

Mixtures (memory) ..... **FULL RICH**

Propellers (memory) ..... **FULL FORWARD**

Throttles (memory) ..... **FULL (42 in Hg max)**

Flaps (memory) ..... **UP**

Airspeed (memory) ..... **100 KIAS (Vyse) – never below 74 KIAS**

▶ Landing gear position?

Gear DOWN:

With gear down single-engine climb is marginal or negative at most weights. Retract if committed to fly; otherwise land straight ahead.

Landing gear (memory) ..... **UP (if committed to fly)**

Inoperative engine (memory) ..... **IDENTIFY – dead foot, dead engine**

Inoperative engine throttle (memory) ..... **VERIFY – retard, no change**

Inoperative engine propeller (memory) ..... **FEATHER**

Inoperative engine mixture (memory) ..... **IDLE CUTOFF**

Gear UP:

Inoperative engine (memory) ..... **IDENTIFY – dead foot, dead engine**

Inoperative engine throttle (memory) ..... **VERIFY – retard, no change**

Inoperative engine propeller (memory) ..... **FEATHER**

Inoperative engine mixture (memory) ..... **IDLE CUTOFF**

Bank ..... **2–5° toward operating engine, ball half out**

Trim ..... **RUDDER & AILERON for straight flight**

Climb ..... **100 KIAS to safe altitude, then Engine Securing**

Deferred: One engine inoperative after takeoff – single-engine approach and landing, no go-around once committed. ..... **NOTED**

Engine Failure in Flight

Loss of power on one engine at altitude

LAND AS SOON AS POSSIBLE

Mixtures (memory)	**FULL RICH**
Propellers (memory)	**FULL FORWARD**
Throttles (memory)	**FULL (as required, 42 in Hg max)**
Landing gear / flaps (memory)	**UP**
Airspeed (memory)	**100 KIAS (Vyse) minimum**
Inoperative engine (memory)	**IDENTIFY – dead foot, dead engine**
Inoperative engine throttle (memory)	**VERIFY – retard, no change**
Inoperative engine propeller (memory)	**FEATHER**
Inoperative engine mixture (memory)	**IDLE CUTOFF**
Bank / rudder	**2–5° toward operating engine, ball half out**
Trim	**RUDDER & AILERON**

▶ Cause obvious (fuel / tank empty) and altitude available for a restart?

Yes – attempt air start:

Fuel selector	**INBOARD (or tank with fuel), electric fuel pump ON**

Use the Air Start procedure. If unsuccessful, secure the engine.

No – secure engine:

Continue with Engine Securing, then divert to the nearest suitable airport. Single-engine service ceiling is low – check drift-down altitude against terrain.

Deferred: One engine inoperative – single-engine approach and landing, drift-down altitude check, no go-around once committed. **NOTED**

Engine Securing (after feather)

Inoperative engine propeller feathered

Inoperative engine fuel selector	**OFF**
Inoperative engine electric fuel pump	**OFF**
Inoperative engine magnetos	**OFF**
Inoperative engine alternator	**OFF**
Inoperative engine cowl flap	**CLOSED**
Propeller synchrophaser	**OFF (if fitted)**
Electrical load	**REDUCE (single alternator)**
Operating engine	**MONITOR – power as required (42 in Hg max), cowl flap as required, mixture adjusted, TIT below 1650°F**
Crossfeed	**ON as required to use fuel from the dead-engine side (level flight only; both electric pumps as required)**
Trim	**ADJUST for straight flight**
ATC	**ADVISE – nearest suitable airport**

Hydraulic pressure for gear and flaps is available from the operating engine pump; expect slower gear and flap operation.

Air Start (Unfeather)

Restart attempt with altitude and cause identified

Fuel selector	**INBOARD (tank with fuel)**
Electric fuel pump	**ON**
Throttle	**OPEN approx 1/4**
Mixture	**FULL RICH**
Magnetos	**ON (both)**
Alternator	**OFF until running**
Propeller	**FORWARD out of feather detent to mid-range (accumulator unfeathers if fitted – allow 10–20 s)**
Starter	**ENGAGE if propeller does not windmill (airspeed 100–120 KIAS helps)**

▶ Engine restarted?

Yes:

Throttle	**1000–1500 RPM until oil pressure and temperature normal, then increase power SLOWLY (cold turbocharger)**
Alternator	**ON**
Electric fuel pump	**OFF when fuel flow steady**
Mixture / cowl flap	**AS REQUIRED**

Deferred: Engine restarted in flight – land as soon as practical, cause investigation required. **NOTED**

No:

Propeller	**FEATHER**
Mixture	**IDLE CUTOFF**

Complete Engine Securing.

Single-Engine Approach, Landing & Go-Around

One engine secured

Operating engine fuel selector	**INBOARD (crossfeed as required until final approach, then OFF)**
Operating engine electric fuel pump	**ON**
Mixture	**FULL RICH**
Propeller	**FULL FORWARD**
Cowl flap	**AS REQUIRED**
Airspeed	**100 KIAS (Vyse) until landing assured**
Approach	**NORMAL glidepath or slightly high – avoid low, flat approaches and large power changes**
Landing gear	**DOWN when landing assured (within gliding distance) – allow for slower hydraulic extension**
Flaps	**15° when landing assured; 40° on short final only**
Airspeed	**REDUCE below 100 KIAS only on short final**

▶ Go-around required?

No:

Landing	**NORMAL – brakes as required, expect yaw as power is reduced**

Yes – above decision height:

Operating engine	**FULL POWER (smoothly, 42 in Hg max)**
Flaps	**UP**
Landing gear	**UP (positive climb)**
Airspeed	**100 KIAS (Vyse)**
Bank	**2–5° toward operating engine**

A single-engine go-around below approx 300 ft AGL, or once 40° flap is extended, should not be attempted.

Engine Fire in Flight

LAND IMMEDIATELY

Affected engine fuel selector (memory)	**OFF**
Affected engine throttle (memory)	**CLOSED**
Affected engine propeller (memory)	**FEATHER**
Affected engine mixture (memory)	**IDLE CUTOFF**
Affected engine electric fuel pump (memory)	**OFF**
Cabin heater / defroster	**OFF**
Affected engine magnetos	**OFF**
Affected engine alternator	**OFF**
Affected engine cowl flap	**CLOSED**
Crossfeed	**OFF**
Airspeed	**100 KIAS or higher to help extinguish fire (descend if required)**
Landing	**NEAREST SUITABLE AIRPORT – do not attempt restart**

Engine Fire During Start

Starter (memory)	**CONTINUE CRANKING**

▶ Engine starts?

Yes:

Throttle	**1500–1700 RPM for a few minutes**
Engine	**SHUT DOWN and inspect for damage**

No:

Mixture	**IDLE CUTOFF**
Throttle	**FULL OPEN**
Electric fuel pump	**OFF**
Fuel selector	**OFF**
Starter	**CONTINUE CRANKING to draw fire into engine**
Magnetos / battery master / alternators	**OFF**
Fire extinguisher	**OBTAIN / USE**
Aircraft	**EVACUATE**

Electrical / Cabin Fire

LAND AS SOON AS POSSIBLE

- Battery master / alternator switches (memory) ..... **OFF**
- Avionics master (memory) ..... **OFF**
- Cabin heater (memory) ..... **OFF**
- Cabin vents / air ..... **CLOSED until fire out, then OPEN to ventilate**
- Fire extinguisher ..... **ACTIVATE (then ventilate cabin)**

▶ Fire out and electrical power required?

Yes:

- Circuit breakers ..... **CHECK – do not reset faulted**
- Battery master ..... **ON**
- Alternators / avionics / equipment ..... **ON one at a time, check for smoke**
- Deferred: Electrical fire – suspect circuits isolated, affected systems unavailable. ..... **NOTED**

No:

Gear and flaps are hydraulic and remain available. Gear position lights, fuel quantity, electric fuel pumps and gear warning are electric – confirm gear by hydraulic pressure cycle and visual check.

- Deferred: Electrical power off – no gear lights or electric fuel pumps; land on inboard tanks, longer runway. ..... **NOTED**

Emergency Descent

- Throttles (memory) ..... **CLOSED (slowly enough to avoid turbo surge; some MP helps cooling)**
- Propellers (memory) ..... **FULL FORWARD**
- Landing gear (memory) ..... **DOWN (below 156 KIAS)**
- Flaps (memory) ..... **UP**
- Airspeed (memory) ..... **156 KIAS (Vle) – smooth air; Vno 180 KIAS gear up if structural cooling allows**
- Mixtures ..... **ENRICH as required**
- Cowl flaps ..... **CLOSED**
- Level-off ..... **POWER UP GRADUALLY – avoid rapid throttle advance**

Emergency Gear Extension

Gear does not extend with the selector; hydraulic pressure low or lost

- Airspeed ..... **100 KIAS or less (slower is better)**
- Landing gear selector ..... **DOWN**
- Hydraulic pressure ..... **CHECK – if available, operating-engine pump may still extend gear; allow time**
- Emergency gear extension lever / knob ..... **PULL and HOLD to the stop (floor, forward of pilot seat / between seats – releases uplocks; CO2 bottle assists on early aircraft)**
- Hand pump ..... **OPERATE until three green (hand-pump variant, if fitted)**
- Aircraft ..... **YAW gently side to side to help gear lock down**
- Gear lights ..... **THREE GREEN**

Do not move the gear selector up or attempt to retract after an emergency extension. Leave the gear down.

▶ Three green obtained?

Yes:

- Deferred: Gear extended by emergency system – do not retract; gear and hydraulic inspection required. ..... **NOTED**

No:

Continue to Gear-Up / Partial Gear Landing procedure.

Gear-Up / Partial Gear Landing

Gear cannot be confirmed down

Runway	**LONGEST HARD SURFACE available, emergency services advised**
Fuel	**BURN DOWN to minimum practical, land on inboard tanks**
Passenger briefing	**COMPLETE (belts tight, brace, airstair door and overwing exit)**
Landing gear	**UP (or as it is – do not cycle)**
Flaps	**40°**
Approach	**NORMAL, 90–95 KIAS**
Electric fuel pumps	**OFF**
Airstair door	**UNLATCHED on short final**

On short final:

Mixtures	**IDLE CUTOFF**
Fuel selectors	**OFF**
Magnetos	**OFF**
Battery master / alternators	**OFF**
Touchdown	**MINIMUM SPEED, slightly nose high, wings level**
Aircraft	**EVACUATE once stopped**

Prototype content. Verify against the operator's approved checklist and the POH/AFM before any use. AeroE Checklist 38b19394 · 2026-11-07